Signal Evaluation & CAPA-need Evidence

Product code **LLZ** — System, Image Processing, Radiological · LLZ reports suddenly increased (IntelliSpace Cardiovascular reports include error 500)

DRAFT — REQUIRES HUMAN REVIEW — prepared by software; not approved

> **Whether a CAPA is required is decided by the company's own CAPA-need form. This document is evidence only.** MaudeSignal never decides whether a CAPA is needed, whether an event is reportable, or which devices are similar.

Document control

Document ID	**SER-LLZ-20260831-001**	**Revision**	0.2
Status	Draft — CAPA-need evidence pack awaiting QA/RA review	**Generated (UTC)**	2026-10-31 03:29:27
Effective date	Not effective — not approved	**CAPA ID**	CAPA-DRAFT-CB63076E
Signal as-of	2026-08-31	**Source data**	openFDA MAUDE device events · demo.db (latest openFDA date_received 20260831)

Device

Product code / classification	LLZ — System, Image Processing, Radiological (21 CFR 892.2050, Class 2, per openFDA)
Device name(s) in reports	IntelliSpace Cardiovascular (spellings that differ only in capitals are grouped)
510(k) number(s) in reports	K153022 (openFDA `pma_pmn_number`)
Manufacturer(s)	Philips Medical Systems Nederland B.V., Best, NL

Approvals (signature manifestation modeled on 21 CFR 11.50: printed name, date/time, meaning)

Role	Printed name	Date / time (UTC)	Meaning	Audit reference
Prepared by	`demo-restore`	2026-10-28 20:36:53	Authorship (software draft)	capa.draft entry in Appendix **Restored from demo/transcripts/20261028T203653Z-05-capa-need-evidence.json** (recorded 2026-10-28T20:36:53Z): a replay of a stored copilot run, written by the restore step, not by the model at this time.
QA/RA reviewer	_____	_____	Approved / Rejected	—
CAPA owner (acceptance)	_____	_____	Responsibility	—

> **Not a 21 CFR Part 11 signature.** The reviewer name is recorded by the Actions page without authenticated identity (no 11.100/11.200/11.300 controls, no 11.70 signature-to-record linking, system not validated per 11.10(a)). The block is *modeled on* 11.50 so a human-readable record shows who decided, when, and what the decision meant; a manufacturer must re-execute approval in its validated eQMS.

> **Number integrity.** 16 distinct figures printed in this record were recomputed from stored openFDA records by the claim-level gate at generation: 16 matched, 0 did not. Each figure carries its claim in the source HTML (`data-claim`) for independent re-verification.

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2. Signal evaluation

2.1 Method

Deterministic detector `maudesignal.monitor.deterioration` over stored openFDA MAUDE reports, anchored on FDA *date_received*. Current window: the 7 days ending on the as-of date (2026-08-25 to 2026-08-31). Baseline: the 4 weeks before it (2026-07-28 to 2026-08-24). Rules (several may fire; precedence: new kind of problem, then more serious than usual, then reports suddenly increased):

Reports suddenly increased	current-window count ≥ 2× the baseline weekly mean **and** ≥ 3 reports
More serious than usual	Injury + Death share of the current window exceeds the baseline share by ≥ 15 percentage points (baseline non-empty)
New kind of problem	first-ever reports for the product code, or an event type never seen before the current window

2.2 Result: REPORTS SUDDENLY INCREASED (rules fired: reports suddenly increased)

Metric	Current window	Baseline (4 weeks)
Reports received	**33**	**35**
Weekly mean (baseline ÷ 4)	—	**8.75**
Sudden-increase threshold (2 × weekly mean)	**17.50**	—
Ratio current ÷ weekly mean	**3.77×**	—
Injury + Death reports	**0**	**1**
Injury + Death share	**0.0%**	**2.9%**
Event type “Malfunction” (current window)	**33**	—

2.3 Weekly report counts

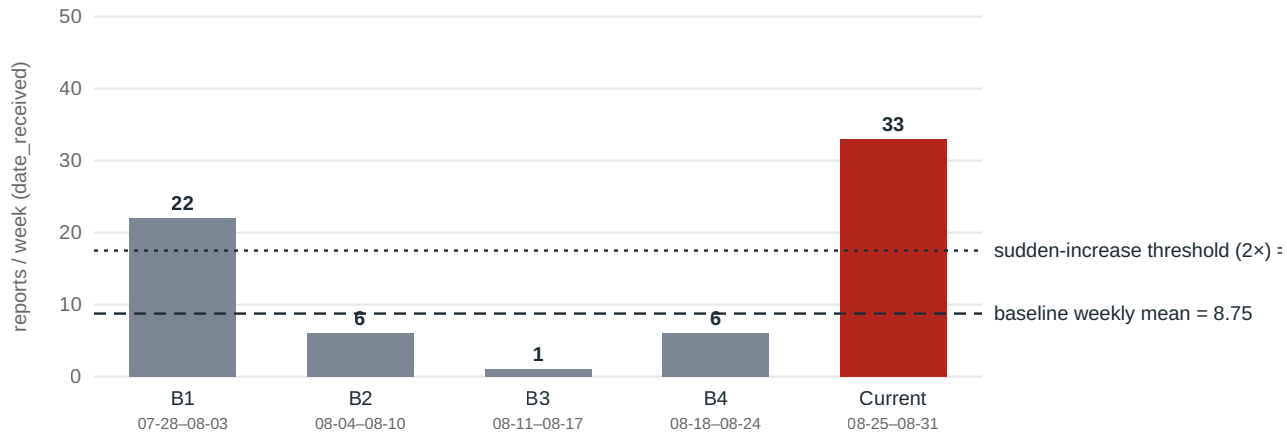


Figure 1 — Reports per week by FDA date_received (B1–B4 = baseline weeks). Deterministic chart drawn from the stored records; not an AI-generated image.

> A sudden increase in reports is a statistical signal for human evaluation. It does not establish a device nonconformity, a root cause, or MDR reportability.

3. Evidence — supporting MAUDE reports

3 reports cited by CAPA-DRAFT-CB63076E. Narrative excerpts are verbatim from openFDA (first 300 characters, whitespace collapsed). Report IDs link to the FDA MAUDE record.

#	MAUDE report	Mfr report no.	Received	Event type	Brand	Narrative excerpt (first 300 chars)
1	50426770	3003768277-2026-101324	2026-08-25	Malfunction	IntelliSpace Cardiovascular	Philips received a complaint regarding IntelliSpace Cardiovascular (ISCV), indicating that they have a case that needs to be corrected. The device was in clinical use at the time of the event, and no adverse event or patient harm was reported (PR (b)(4)).Philips has started an investigation of this…
2	50426773	3003768277-2026-101323	2026-08-25	Malfunction	IntelliSpace Cardiovascular	Philips received a complaint regarding IntelliSpace Cardiovascular (ISCV), indicating that unable to finalize report on Echo Module and encounter an error 500. The device was in clinical use at the time of the event, and no adverse event or patient harm was reported (PR (b)(4)).Philips has started…
3	50448213	3003768277-2026-101328	2026-08-26	Malfunction	IntelliSpace Cardiovascular	Philips received a complaint regarding IntelliSpace Cardiovascular (ISCV), indicating that an error 500 occurs for one study, preventing the doctor from processing the report. The device was in clinical use at the time of the event, and no adverse event or patient harm was reported (PR (b)(4)).Phil…

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4. CAPA-need evidence pack and blank CAPA sections for the CAPA owner (ISO 13485:2016 §8.5.2 / §8.5.3, incorporated by 21 CFR Part 820 — QMSR)

Whether a CAPA is required is decided by the company's own CAPA-need form. This document is evidence only. The sections below are left blank for the CAPA owner if their form decides a CAPA is needed.

Evidence record ID	CAPA-DRAFT-CB63076E	**Status**	DRAFT
Title	LLZ reports suddenly increased (IntelliSpace Cardiovascular reports include error 500)		
Opened (UTC)	2026-10-28 20:36:53	**Opened by**	`demo-restore`
Source	Postmarket surveillance signal SER-LLZ-20260831-001 (§2) · complaint/MDR data (§3) · MDR worksheet (§5)		

Under the QMSR (effective 2026-02-02) CAPA requirements are those of ISO 13485:2016 §8.5.2 and §8.5.3; the former 21 CFR 820.100 is superseded and is not cited.

4.1 (a) Review of nonconformities, including complaints

Problem statement as prepared (demo-restore, recorded verbatim from the CAPA-need evidence pack):

> LLZ reports suddenly increased this week with IntelliSpace Cardiovascular malfunction reports describing error 500 preventing report finalization and processing

Reports that contain “error 500” in their stored text: 50426773, 50448213 (2 of 3 cited reports). Cited in this pack but not for that phrase: 50426770. The weekly counts in §2 cover every report of product code LLZ, not only those that contain the phrase.

> **Gate:** the drafted text contains figures or identifiers without a structured claim — treat as unverified: '500 preventing report' has no count claim

Complaint/MDR input reviewed: **3** MAUDE reports (§3); signal result **reports suddenly increased** (§2).

4.2 (b) Determination of the cause(s) of nonconformity

MaudeSignal does not determine root cause.

To be completed by CAPA owner

_____

_____

_____

_____

Name: _____ Date (UTC): _____

4.3 (c) Evaluation of the need for action to ensure nonconformities do not recur

Input: deterministic rule fired — reports suddenly increased. Need-for-action decision and risk assessment rationale:

To be completed by CAPA owner

_____

_____

_____

Name: _____ Date (UTC): _____

4.4 (d) Planning, documenting and implementing action, including updating documentation

To be completed by CAPA owner

Name: _____ Date (UTC): _____

4.5 (e) Verification that the action does not adversely affect the ability to meet regulatory requirements or the safety and performance of the device

To be completed by CAPA owner

Name: _____ Date (UTC): _____

4.6 (f) Review of the effectiveness of the corrective action

Effectiveness criteria, monitoring period and result:

To be completed by CAPA owner

Name: _____ Date (UTC): _____

4.7 Records of results of investigation and action (§8.5.2, final paragraph)

This document (revision 0.2, shown in document control) and the audit trail in Appendix A. Reporting to management review: pending.

4.8 Preventive action (§8.5.3) — potential nonconformities in other device types you choose (product codes)

To be completed by CAPA owner

Name: _____ Date (UTC): _____

4.9 Complaint handling link (§8.2.2; 21 CFR 820.35(a))

Complaint records for events reportable under Part 803 must hold device name, date received, UDI/identifiers, complainant contact, nature and details, correction/corrective action and any reply. MaudeSignal supplies the public MAUDE view only; link the manufacturer's complaint file numbers here:

To be completed by CAPA owner

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5. MDR assessment worksheet (FDA Form 3500A map)

DRAFT WORKSHEET — NOT FOR SUBMISSION

Reportability is a human regulatory decision.

5.1 Reporting clock (advisory — 21 CFR 803.50 / 803.53)

Awareness date (803.3 / 803.50)	**2026-10-28** (assumed = CAPA-need evidence pack creation date (demo); confirm)
30-calendar-day report due (803.50(a))	2026-11-27
5-work-day report due, if triggered (803.53(a) remedial action / (b) FDA request)	2026-11-04

Dates computed by `copilot.tools.deadline_check` (same code as the copilot). A clock only — it does not decide that any event is reportable.

5.2 Reportability determination (human)

☐	803.50(a)(1): information reasonably suggests the device may have caused or contributed to a death or serious injury
☐	803.50(a)(2): the device malfunctioned and would be likely to cause or contribute to a death or serious injury if the malfunction recurred
☐	803.53(a): remedial action needed to prevent an unreasonable risk of substantial harm (5-day report) · ☐ 803.53(b): FDA written request
☐	Not reportable — rationale documented in the MDR event file (803.18(b)(1)(i))

To be completed by CAPA owner

_____

_____

Name: _____ Date (UTC): _____

5.3 Form 3500A fields — pre-filled from openFDA where available

One 3500A is filed per reportable event. Pre-fills are the values common to the cited reports, from openFDA; blank cells require manufacturer information (21 CFR 803.52: “if known or reasonably known”).

Sec.	Item	Field	803.52	Pre-fill	Source
A	A1–A5	Patient identifier, age, sex, weight, race/ethnicity	803.52(a)	Not pre-filled — MaudeSignal handles no patient-identifying information	—
B	B1	Type of report (adverse event / product problem …)	803.52(b)	_____	—
B	B2	Outcomes attributed to adverse event	803.52(b)	_____	—
B	B3	Date of event	803.52(b)	Per report — not in the evidence table; open each FDA record (date received is shown)	—
B	B5	Describe event or problem	803.52(b)	Per report — MAUDE narrative excerpts in Evidence table	mdr_text
B	B6–B7	Relevant tests/laboratory data; other relevant history	803.52(b)	_____	—
D	D1	Brand name	803.52(c)	IntelliSpace Cardiovascular	openFDA device event: device.brand_name
D	D2a	Product code (procode)	803.52(c)	LLZ	openFDA device event: device.device_report_product_code

DRAFT WORKSHEET NOT FOR SUBMISSION

Sec.	Item	Field	803.52	Pre-fill	Source
D	D2b	Common device name	803.52(c)	Cardiovascular Image and Information Management System	openFDA device event: device.generic_name
D	D3	Manufacturer name, city, state, country	803.52(c)	Philips Medical Systems Nederland B.V., Best, NL	openFDA device event: device.manufacturer_d_*
D	D4	Model #, lot #, catalog #, serial #, UDI	803.52(c)	Model: IntelliSpace Cardiovascular	openFDA device event: device.model_number
D	D5–D10	Operator, implant/explant dates, reprocessing, 3rd-party service, availability for evaluation, concomitant products	803.52(c)	_____	—
E	E1–E4	Initial reporter (name, address, health professional, occupation)	803.52(d)	_____	—
F	F1–F13	User facility / importer block	803.52(f) (11)	Not applicable to a manufacturer report unless correcting UF/importer data	—
G	G1	Contact office and manufacturing site	803.52(e)	_____	—
G	G2	Report source	803.52(e)	_____	—
G	G3	Date received by manufacturer (awareness)	803.52(e)	See awareness date above (human entry)	user input
G	G4	PMA / 510(k) number	803.52(e)	K153022	openFDA device event: pma_pmn_number
G	G6	Type of report (initial / follow-up / 5-day …)	803.52(e)	_____	—
G	G7	Adverse event term(s)	803.52(e)	_____	—
G	G8	Manufacturer report number	803.52(e)	_____	—
H	H1	Type of reportable event (death / serious injury / malfunction)	803.52(f)	openFDA event_type observed: Malfunction	openFDA device event: event_type (reference only)
H	H3	Device evaluated by manufacturer? (summary or reason not)	803.52(f)	_____	—
H	H6	Adverse event problem codes (device, patient, evaluation)	803.52(f)	_____	—
H	H7	Remedial action initiated (type)	803.52(f)	_____	—
H	H9	Correction/removal report number (21 USC 360i(g))	803.52(f)	_____	—
H	H10	Additional manufacturer narrative	803.52(f)	_____	—
—	FDA	Date received by FDA (MAUDE)	—	2026-08-25 to 2026-08-26	openFDA device event: date_received

MDR procedures and event files: 21 CFR 803.17 (written procedures) and 803.18 (event files incl. the reportability deliberation; retain the longer of 2 years or the expected device life).

DRAFT WORKSHEET NOT FOR SUBMISSION

6. Narrative summary

> AI-drafted (live run, 2026-10-31T03:30:21Z). Drafted by muse-spark-1.3 (skill dossier-narrative v1.2.0); every sentence was re-checked by the deterministic claim-level gate against stored openFDA records. Withheld sentences are shown struck through with the gate's reason. Shown: 5 · withheld: 0.

VERIFIED The monitor noted that reports suddenly increased, with 33 reports in the current window compared with 35 in the baseline period.
VERIFIED A cited report stated "they have a case that needs to be corrected" and noted that investigation had started.
VERIFIED A cited report described "unable to finalize report on Echo Module and encounter an error" during clinical use.
VERIFIED The cited reports named IntelliSpace Cardiovascular as the brand.
VERIFIED Root cause, MDR reportability decision, and effectiveness check remained open for the CAPA owner and the regulatory reviewer.

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Appendix A. Audit trail

Entries of the append-only, SHA-256 hash-chained `action_audit` table for CAPA-DRAFT-CB63076E and SER-LLZ-20260831-001. Time stamps are computer-generated (UTC).

Seq	Time (UTC)	Actor	Action	Entity	Entry hash (prefix)
1	2026-10-31 03:29:27	`demo-restore`	`capa.draft`	`CAPA-DRAFT-CB63076E`	`d286cef57220`

> The `capa.draft` entry is a replay: restored from demo/transcripts/20261028T203653Z-05-capa-need-evidence.json (recorded 2026-10-28T20:36:53Z), written under the actor `demo-restore`. Its table time is the replay time; the recording time is in the entry payload.

> **Chain verification** (`Database.verify_audit_chain()`, all entries recomputed): INTACT.

> Modeled on 21 CFR 11.10(e) (secure, computer-generated, time-stamped audit trails). Not claimed as Part 11 compliant: the system is not validated, has no user authentication, and a database administrator could rewrite the whole chain consistently; the hash chain detects edits to individual entries only.

Generated by MaudeSignal 0.1.0 (maudesignal.dossier). Regulatory sources: docs/dossier_sources.md.

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Appendix B. Search methodology

Every search behind this pack, so a reviewer can repeat it. Rows marked `dossier:` are the queries this document ran over the stored reports; the other rows are the copilot tool calls stored in the transcript(s) whose recorded CAPA id is CAPA-DRAFT-CB63076E. Dates are FDA *date_received* (YYYYMMDD). “not recorded” means the stored log did not keep that value; nothing is estimated. No model call produced this table.

#	Search	Dataset	Filters	From	To	Count	Run at (UTC)
1	`dossier: current window`	openFDA MAUDE reports (stored copy)	`product_code=LLZ`	20260825	20260831	33	`2026-10-31T03:29:27Z`
2	`dossier: baseline window`	openFDA MAUDE reports (stored copy)	`product_code=LLZ`	20260728	20260824	35	`2026-10-31T03:29:27Z`
3	`dossier: event type`	openFDA MAUDE reports (stored copy)	`event_type=malfunction; product_code=LLZ`	20260825	20260831	33	`2026-10-31T03:29:27Z`
4	`dossier: cited reports`	openFDA MAUDE reports (stored copy)	`product_code=LLZ; report_ids=50426770,50426773,50448213`	—	—	3	`2026-10-31T03:29:27Z`
5	`weekly_signals`	openFDA MAUDE reports (stored copy)	`none`	20260825	20260831	not recorded	`2026-10-28T20:36:53Z`
6	`get_report`	openFDA MAUDE reports (stored copy)	`report_id=50426770`	—	—	1	`2026-10-28T20:36:53Z`
7	`get_report`	openFDA MAUDE reports (stored copy)	`report_id=50426773`	—	—	1	`2026-10-28T20:36:53Z`
8	`get_report`	openFDA MAUDE reports (stored copy)	`report_id=50448213`	—	—	1	`2026-10-28T20:36:53Z`

> Copilot rows carry the time the transcript was recorded; individual calls are not time-stamped. Counts for copilot rows are read from the stored result preview.

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